INTERMEDIATE-CARDINALITY STRATA AND DIRECT LIMITS IN LEAN: A FORMALIZATION STUDY AROUND THE CONTINUUM

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Abstract. This paper draft presents a Lean 4 formalization study of an Intermediate-Cardinality Arithmetic Hypothesis (ICAH): a stratified view of the real continuum under the negation of the Continuum Hypothesis. The development packages intermediate-size subsets of $\mathbb{R}$ as strata, refines strata that are closed under field operations as subfield strata, bundles cardinal data with ordered-field structure through size-aware fields, and studies elementary chains and their direct limits in the language of rings. The current formalization contains several fully proved components, including an intermediate-cardinal witness under $\neg$ CH, a concrete synthetic stratum of cardinality $\aleph_1$, a countability bound for the real algebraic numbers, definability lemmas for the graphs of addition and multiplication, and a cardinal-arithmetic theorem for direct limits. The remaining dependencies are deliberately exposed as named axioms, yielding both a mathematical statement and a roadmap for Mathlib contributions.

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1. Introduction

The Continuum Hypothesis (CH) asserts that there is no cardinal strictly between the cardinality of the natural numbers and the cardinality of the continuum. Consequently, any mathematical program that treats intermediate cardinalities inside $\mathbb{R}$ must either work relative to $\neg$ CH, change the meaning of size, or explicitly track which assumptions are external to ZFC.

The Lean development studied here chooses the first route. It introduces a named set-theoretic assumption, `ICAH.not_CH : continuum ≠ aleph 1`, and uses it to witness an intermediate cardinal. Around this assumption, the project builds a formal architecture for strata of the real continuum, size-aware field structures, subfield refinements, definability kernels, elementary chains, and direct limits.

Contribution. The main contribution of the current draft is not a new set-theoretic independence theorem. It is a formalization architecture: a Lean-readable decomposition of a continuum-stratification hypothesis into proved lemmas, named assumptions, and Mathlib-facing gaps.

The project should be positioned for a mathematical audience as a formalization paper with three layers:

1. a set-theoretic layer, where $\neg$ CH produces intermediate cardinal witnesses;
2. an algebra/model-theoretic layer, where subfields of $\mathbb{R}$ carry field and order structure and are intended to support real-closed-field semantics;

3. a proof-engineering layer, where every remaining dependency is exposed as a named axiom and can be attacked as an independent Mathlib contribution.

The present paper draft is organized as follows. Section 2 recalls the mathematical background. Section 3 describes the Lean architecture. Section 4 isolates the results that are already proved. Section 5 gives the axiom inventory and explains which parts are mathematical assumptions and which parts are formalization gaps. Section 6 discusses related work. Section 7 sketches analogues in valued and p-adic field theory.

1.1. **Guiding problem.**

Definition. An *intermediate-cardinality stratum* is, informally, a subset $S \subseteq \mathbb{R}$ equipped with a cardinal κ such that

$$\aleph_0 < \kappa < 2^{\aleph}_0$$

and a proof that the subtype determined by S has cardinality κ.

Definition. A *size-aware field* is a bundled field-like object carrying its underlying type, a designated cardinal, and the typeclass instances needed to treat the carrier as an ordered ring or field in Lean.

The central formal question is:

Question. Can one build, under explicit assumptions, a hierarchy of intermediate-size strata whose carriers support enough algebra and model theory to form elementary chains with a direct limit of cardinality continuum?

The current Lean code answers part of this question constructively and isolates the rest as a small collection of named assumptions.

2. Background

2.1. **The continuum and the role of ¬ CH.**

Let $c = 2^{\aleph}_0$ denote the cardinality of the continuum. The Continuum Hypothesis states that there is no cardinal κ satisfying $\aleph_0 < \kappa < c$. Since Gödel and Cohen, the standard mathematical background is that CH is independent of ZFC, assuming ZFC is consistent [1, 2].

The Lean project therefore does not attempt to prove ¬ CH in ZFC. Instead, it introduces ¬ CH as a named axiom and develops all downstream objects relative to that assumption.

Proposition. Under the project axiom `ICAH.not_CH`, the cardinal $\aleph_1$ witnesses the existence of an intermediate cardinal:

$$\exists \kappa, \aleph_0 < \kappa \wedge \kappa < 2^{\aleph}_0.$$

The Lean proof uses the standard inequality $\aleph_0 < \aleph_1$, the theorem $\aleph_1 \leq c$, and converts the inequality plus non-equality $c \neq \aleph_1$ into the strict inequality $\aleph_1 < c$.

2.2. **Real closed fields and definability.**

Real closed fields are the algebraic and model-theoretic setting closest to the ordered real field. They admit quantifier-elimination results and provide the natural language in which order, addition, multiplication, and definability can be studied together [3, 4].

ICAH uses this background in a restrained way. The code does not merely assume every bare subset of $\mathbb{R}$ is a field. Instead, it introduces `SubfieldStratum`, which requires the carrier set to be the underlying set of a `Subfield RR`. This is the right formal move: closure under field operations is obtained from the subfield structure rather than reconstructed ad hoc.

2.3. **Elementary chains and direct limits.**

An elementary chain is a sequence of structures connected by elementary embeddings. The classical model-theoretic expectation is that unions or directed limits of elementary chains preserve the relevant first-order theory under suitable hypotheses [5].

The Lean project packages this through `ElemChain`, where each level carries an `LOR`-structure and each successor map is an elementary embedding. The direct limit is then implemented using Mathlib's `Language.DirectLimit` API.

3. Formal architecture

This section records the paper-level interpretation of the Lean objects. The goal is to make the repository legible to mathematicians without forcing them to read Lean code first.

3.1. **Core objects.**

Definition. `Stratum` is the basic continuum-layer object. It consists of an ordinal index, a set $S \subseteq \mathbb{R}$, a cardinal κ, a witness that the subtype determined by S has cardinality κ, and bounds $\aleph_0 < \kappa < c$.

Definition. `SizeAwareField` is a bundled structure containing a carrier type, a designated cardinal, and the instances `[Field K]`, `[LinearOrder K]`, and `[IsStrictOrderedRing K]`. This avoids relying on deprecated or overly strong ordered-field classes while still exposing enough structure for model-theoretic use.

Definition. `SubfieldStratum` refines `Stratum` by adding a subfield `subfield : Subfield RR` and a proof that the stratum set is exactly the underlying set of that subfield.

The role of `SubfieldStratum` is central. A raw set of real numbers is not automatically closed under addition, multiplication, negation, and inverses. A subfield is. Thus, the field part of the theory is moved from a fragile closure proof obligation into a stable algebraic structure.

Construction. `subfieldToSAF` converts a subfield of $\mathbb{R}$, together with a cardinality witness, into a `SizeAwareField`. The key design choice is to use subtype inheritance for the linear order and Mathlib's subfield ordered-ring instance for the strict ordered ring structure.

3.2. **Definability layer.**

The formalization uses an abbreviation `LOR` for the first-order language used in the definability kernel. In the current development, the graphs of addition and multiplication on $\mathbb{R}$ are proved definable by constructing bounded formulas and transporting realization through Mathlib's language-homomorphism machinery.

Remark. The definability results are important because they demonstrate that the paper is not only doing cardinal bookkeeping. It also touches the first-order model-theoretic infrastructure needed to express arithmetic inside a layer.

3.3. **Elementary chains.**

Definition. `ElemChain` packages a sequence `obj : NN -> Type*` of first-order structures and elementary embeddings `obj n ↪ₑ obj (n+1)`.

From these successor maps, the project defines two related systems:

1. `embLE`, which composes successor elementary embeddings and preserves elementarity;
2. `sysEmb`, which uses Mathlib's directed-system API to build the underlying embedding system.

The lemma `embLE_eq_sysEmb` proves that these two constructions agree as functions. This is a small but important bridge: the proof-relevant elementary embedding API and the directed-colimit API are not automatically the same object.

3.4. **Direct limit.**

Definition. For an elementary chain `C`, the direct limit `DirectLim C` is defined as `Language.DirectLimit C.obj (...)`, using the underlying directed system of embeddings.

The direct limit is the formal version of the intended limit field F_ω. The project currently proves a substantial cardinal theorem for this direct limit and exposes the elementarity of the direct limit as a named Mathlib gap.

4. Proved results

This section separates fully proved Lean results from assumptions or future work. In the final paper, each result should include a pointer to the Lean declaration and, when appropriate, a compressed proof sketch.

4.1. **Intermediate cardinal witness.**

Theorem. `exists_intermediate_cardinal`: assuming `ICAH.not_CH`, there exists a cardinal κ such that $\aleph_0 < \kappa < c$.

Remark. The witness is $\aleph_1$. The proof uses `aleph0_lt_aleph_one`, `aleph_one_le_continuum`, and turns non-equality into strict inequality through `lt_of_le_of_ne`.

4.2. Concrete synthetic stratum.

Construction. `syntheticStratum` materializes an intermediate-cardinality subset of $\mathbb{R}$ by extracting a concrete set $S \subseteq \mathbb{R}$ with cardinality $\aleph_1$ from a cardinal existence theorem.

This is the first bridge from a cardinal statement to an inhabited Lean structure. It turns the abstract existence of an intermediate cardinal into a concrete `Stratum` object.

4.3. Subfield strata and size-aware fields.

Theorem. `fieldOnSubfieldStratum`: every `SubfieldStratum` yields a `SizeAwareField` whose carrier and cardinal match the underlying stratum.

The technical point is that the carrier equality is treated as a propositional equality of subtypes, not merely a bijection. This is precisely what the `SizeAwareField.carrier` field requires.

4.4. Real algebraic numbers as a base field.

Theorem. `algReal_card_le_aleph0`: the real algebraic numbers, represented as `algebraicClosure QQ RR` inside $\mathbb{R}$, have cardinality at most $\aleph_0$.

Together with the lower bound for infinite types, this supports the concrete base object `algRealSAF`, a countable size-aware field outside the intermediate-cardinality hierarchy.

4.5. Definability of addition and multiplication.

Theorem. `graphDefinable_add` and `graphDefinable_mul`: the graphs of addition and multiplication on $\mathbb{R}$ are first-order definable in the selected ring language with parameters.

These lemmas are technically demanding because the proof must build explicit bounded formulas and then verify realization through several layers of Mathlib's model-theory API.

4.6. Elementarity-preserving composition.

Theorem. `embLE`: for any $m \le n$, the chain induces an elementary embedding from level m to level n.

Theorem. `embLE_eq_sysEmb`: the elementarity-preserving embedding built by recursive composition agrees as a function with the directed-system embedding used by `Language.DirectLimit`.

4.7. Cardinality of the direct limit.

Theorem. `directLimit_card`: if every level has cardinality below continuum and the supremum of the level cardinalities is continuum, then the direct limit has cardinality continuum.

The proof has two halves. The upper bound exhibits the direct limit as a quotient of the sigma type $\sum_{n:\mathbb{N}} C_n$, bounding its cardinality by $\aleph_0 \cdot c = c$. The lower bound uses injectivity of the canonical maps from each level into the direct limit and then takes the supremum over levels.

Contribution. For the paper, this is the strongest fully proved mathematical theorem to foreground. It is independent of the grand interpretation of ICAH and can be read as a reusable cardinal-arithmetic theorem for direct limits of countable chains.

5. Axiom inventory and Mathlib roadmap

A central strength of the Lean development is that it does not hide its assumptions. The theorem `icahTheorem` is assembled from proved components plus a small set of named dependencies.

5.1. External mathematical assumption.

Mathlib gap. `ICAH.not_CH`: the negation of the Continuum Hypothesis, represented as `continuum ≠ aleph 1`.

This is not a Mathlib gap. It is the intended set-theoretic regime of the project. The paper should say explicitly that the theory is developed relative to $\neg$ CH.

5.2. Abstract field-on-stratum assumption.

Mathlib gap. `ICAH.fieldOnStratum`: every stratum admits a size-aware field whose carrier and cardinal match the stratum.

This is too strong for arbitrary raw strata. The paper should present `SubfieldStratum` as the corrected refinement and `fieldOnSubfieldStratum` as the proved replacement in the subfield case.

5.3. Intermediate-size subfields.

Mathlib gap. `ICAH.subfieldStratumExists`: under $\neg$ CH, there exists a subfield of $\mathbb{R}$ with cardinality strictly between $\aleph_0$ and c.

The expected classical construction is to take a transcendence basis of $\mathbb{R}$ over $\mathbb{Q}$, choose a subset of size $\aleph_1$, and take the real closure of the generated subfield inside $\mathbb{R}$. The Lean work needed is not the idea itself but the assembly of the relevant field-theoretic and cardinality APIs.

5.4. Real-closedness infrastructure.

Mathlib gap. `ICAH.Real.isRealClosed`: package `IsRealClosed RR` as an available Mathlib fact.

Mathlib gap. `ICAH.subfieldIsRealClosed`: formulate and prove the preservation of real-closedness for the intended elementary subfield setting.

The first gap is likely a direct Mathlib contribution. The second is more structural: it requires connecting `IsRealClosed` to first-order real-closed-field theory or using an appropriate elementary-substructure theorem.

5.5. Direct limits of elementary chains.

Mathlib gap. `ICAH.ElemChain.losDirectLimit`: if every level of an elementary chain embeds compatibly into $\mathbb{R}$, then the direct limit is elementarily equivalent to $\mathbb{R}$.

This is the most important model-theoretic API gap. The expected proof route is to build the underlying embedding by `Language.DirectLimit.lift` and prove it elementary via a Tarski–Vaught argument.

5.6. Recommended order of attack.

1. Polish and submit `IsRealClosed RR` to Mathlib.
2. Extract `directLimit_card` as a reusable theorem or at least document it as a local theorem independent of ICAH.
3. Prove an elementary-direct-limit theorem for `Language.DirectLimit`.
4. Formalize intermediate-cardinality subfields via transcendence bases and real closure.
5. Replace the global `fieldOnStratum` axiom with refined hypotheses, probably using `SubfieldStratum` or a closure predicate.

6. Related work

6.1. Continuum hypothesis and independence.

The set-theoretic background is classical. The Continuum Hypothesis asks whether $2^{\aleph}_0 = \aleph_1$, equivalently whether there is no intermediate cardinal between $\aleph_0$ and the continuum. Gödel proved relative consistency of CH with ZFC, and Cohen proved the independence direction using forcing [1, 2].

The present work does not contribute to the independence problem itself. It works in the $\neg$ CH regime and asks what kind of algebraic and model-theoretic structure can be layered over intermediate-cardinality subsets or subfields of $\mathbb{R}$.

6.2. Real closed fields.

The algebraic and model-theoretic backbone is the theory of real closed fields. Tarski's quantifier elimination for real closed fields makes them a natural setting for definability questions involving addition, multiplication, and order [3, 4].

ICAH uses this tradition, but the novelty of the formalization is to combine real-closed-field ideas with explicit cardinal bookkeeping and Lean-level axiom accounting.

6.3. Formalized mathematics in Lean.

The project belongs to the growing ecosystem of Lean 4 and Mathlib formalizations. Mathlib is a community-driven library of formalized mathematics used for research-scale formal proofs [6]. ICAH is not only a Lean artifact but also a diagnostic of Mathlib boundaries: it identifies concrete places where model theory, real closed fields, cardinal arithmetic, and directed colimits need stronger APIs.

6.4. **Direct limits and elementary chains.**

Elementary-chain arguments are standard in model theory. The formal contribution here is more specific: it tests how far Mathlib's `Language.DirectLimit` API can be pushed toward a fully formal elementary-chain theorem. The direct-limit cardinality result is already proved locally; the elementarity theorem remains a named target.

7. Neighboring fields and analogues

The ICAH architecture has natural analogues beyond real closed fields. The closest comparisons are not arbitrary fields but fields equipped with a closure principle and a first-order theory.

7.1. **P-adically closed fields.**

P-adically closed fields are the valued-field analogue most similar in spirit to real closed fields. On the real side, order and real closure control definability. On the p-adic side, valuation, henselianity, and p-adic closure play the corresponding role. Model theory of valued fields studies exactly this pattern: a field is enriched by extra structure, and the central questions concern definability, elementary equivalence, and preservation under extensions or chains [7, 8].

A p-adic analogue of ICAH would replace:

1. subsets or subfields of $\mathbb{R}$ by valued subfields of a p-adic ambient field;
2. order/real-closure hypotheses by valuation/henselian or p-adically-closed hypotheses;
3. definability in an ordered-ring language by definability in a valued-field language;
4. real-closed-field elementary chains by valued-field elementary chains.

7.2. **Henselian valued fields.**

Henselian valued fields provide a broader comparison class. They are not merely fields with a topology; they are algebraic structures with a valuation satisfying a lifting principle. This makes them suitable for direct analogues of the ICAH pattern: carrier, extra structure, closure property, definability theory, elementary embeddings, and direct limits.

7.3. **Algebraically closed fields.**

Algebraically closed fields are less close to the order-theoretic content of ICAH, but they share an important formal pattern: a strong closure property plus a well-behaved first-order theory. A stripped-down version of the ICAH architecture could be tested in algebraically closed fields as a simpler control case.

7.4. **Paper positioning.**

Remark. The safest comparison is: ICAH is a real-field, cardinal-aware, Lean-formalized instance of a broader model-theoretic pattern. The neighboring fields are real closed fields, p-adically closed fields, henselian valued fields, pseudo real closed fields, pseudo p-adically closed fields, and algebraically closed fields.

For the first paper, these analogies should remain in a future-work section. The core contribution should stay focused on the formal Lean development around the real continuum.

8. Conclusion

The current ICAH formalization is best presented as a rigorous formalization study rather than as a finished foundational theory. Its strongest mathematical components are already useful: the intermediate-cardinal witness under $\neg$ CH, the concrete synthetic stratum, the subfield-to-size-aware-field construction, the definability lemmas for arithmetic graphs, and especially the cardinality theorem for direct limits.

The most important editorial decision is to keep the paper honest about the distinction between proved Lean results and named assumptions. This is not a weakness. It is one of the project's strongest features: the axiom inventory turns an ambitious mathematical program into a precise roadmap of independent formalization problems.

8.1. Immediate next steps.

1. Add a small table mapping each Lean declaration to its mathematical statement and proof status.
2. Verify the current repository build and paste the exact `#print axioms icahTheorem` output into an appendix.
3. Decide whether the first submission target is a formalization venue, a Lean/Mathlib note, or a broader mathematical logic preprint.
4. Promote `directLimit_card` as the headline fully proved theorem.
5. Keep p-adic and valued-field analogies as future work, not as the main thesis.

TODO. Before public release, replace every informal claim about Mathlib gaps with either a direct link to the relevant Mathlib file, a Zulip discussion, or a minimized Lean example showing the missing theorem.

References

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